

RIVERTON, WY, USA

WMO: 726720

Lat: 43.0647N

Lon: 108.4767W

Elev: 5574

StdP: 11.97

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 24061

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.9	-2.3	-13.3	3.3	-5.7	-8.6	4.3	0.3	32.0	36.4	27.7	31.4	3.5	330	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.1	92.5	58.2	90.1	57.8	87.6	57.4	61.7	82.1	60.4	81.1	59.3	80.2	11.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.5	80.7	65.5	53.8	75.8	65.2	51.6	69.9	65.5	30.1	82.0	29.1	81.0	28.2	80.1	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.2	24.5	20.4	-16.6	97.3	6.3	2.3	-21.1	98.9	-24.8	100.3	-28.4	101.6	-33.0	103.3		
			-16.8	64.2	5.8	1.5	-21.0	65.3	-24.4	66.2	-27.7	67.1	-31.9	68.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	20.5	25.0	36.2	44.0	53.3	64.0	73.2	70.1	59.9	45.5	31.9	20.3
	DBStd	20.37	11.41	11.38	10.10	8.56	8.62	7.52	5.00	5.42	8.51	9.70	11.14	11.32
	HDD50	4012	916	700	434	214	66	4	0	0	20	195	544	919
	HDD65	7696	1381	1120	893	630	369	105	6	20	192	604	992	1384
	CDD50	2350	0	0	6	34	170	425	718	623	316	56	2	0
	CDD65	559	0	0	0	0	8	76	259	178	38	0	0	0
	CDH74	7310	0	0	0	10	186	1157	3186	2174	577	20	0	0
	CDH80	3006	0	0	0	1	34	439	1487	882	162	1	0	0
	WSAvg	8.3	6.9	7.7	9.2	10.0	9.3	9.2	8.7	8.4	8.3	8.3	7.3	6.7
	PrecAvg	8.2	0.2	0.3	0.4	1.0	1.6	1.3	0.7	0.5	0.8	0.7	0.4	0.3
(15) Precipitation	PrecMax	13.5	1.5	1.3	3.0	3.7	4.4	5.9	2.9	1.6	4.4	3.6	2.1	3.6
	PrecMin	3.7	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.3	0.3	0.3	0.5	0.7	1.1	1.2	0.6	0.4	0.8	0.7	0.4	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	54.9	68.1	74.8	83.9	93.4	97.3	93.6	89.2	76.9	63.3	52.2
		MCWB	37.0	40.0	44.9	48.5	54.0	56.7	59.0	58.6	57.7	51.3	44.6	38.8
	2%	DB	44.7	49.4	61.9	69.7	79.3	89.6	93.4	90.7	84.3	70.8	56.8	45.3
		MCWB	35.0	37.1	42.4	46.5	52.4	56.6	58.7	57.8	54.9	49.4	42.0	35.3
	5%	DB	40.2	45.1	57.0	64.6	74.6	85.1	91.0	88.3	81.1	65.6	52.3	40.4
		MCWB	32.3	34.6	40.7	44.7	51.5	55.7	58.4	57.4	54.1	47.5	39.7	32.5
	10%	DB	36.5	40.5	52.3	60.8	70.3	81.2	88.4	85.5	76.5	61.3	47.6	36.2
		MCWB	30.0	31.9	38.7	43.2	50.0	54.9	58.0	56.6	52.7	45.4	37.4	29.8
	0.4%	WB	38.6	41.1	45.9	50.2	57.4	61.0	64.1	62.5	60.3	53.2	45.6	39.4
		MCDB	46.6	53.4	64.7	70.9	78.1	82.4	82.7	83.7	79.5	72.5	60.0	50.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	37.9	43.3	47.5	54.4	59.2	62.3	60.9	57.7	50.5	42.4	35.9
		MCDB	43.5	48.0	59.7	65.9	73.2	81.0	82.0	81.6	77.7	67.1	55.7	44.9
	5%	WB	32.8	35.0	41.3	45.8	52.7	57.7	61.1	59.6	55.9	48.4	40.1	32.8
		MCDB	39.9	44.3	55.9	63.0	71.1	78.9	81.6	80.0	74.8	63.6	51.4	40.2
	10%	WB	30.3	32.2	39.1	43.9	51.0	56.3	60.0	58.6	54.3	46.3	37.5	30.0
		MCDB	36.2	40.0	51.4	59.0	67.9	76.4	80.9	79.1	72.6	60.0	46.9	35.6
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.2	23.0	23.6	24.4	27.6	29.1	28.3	26.4	23.0	20.8	18.6
		MCDBR	22.4	24.6	29.0	31.0	31.2	33.4	32.2	31.8	31.6	28.8	26.5	23.0
		MCWBR	15.2	15.1	15.0	14.4	12.3	12.0	11.0	11.5	12.6	13.7	15.1	15.4
	5% WB	MCDBR	21.7	24.3	27.7	28.6	28.6	30.1	28.6	28.5	28.1	27.4	25.7	22.5
		MCWBR	15.1	15.3	14.5	13.7	11.5	11.5	9.7	10.3	11.3	13.5	15.1	15.2
(40) Clear-Sky Solar Irradiance	taub		0.233	0.239	0.258	0.297	0.312	0.314	0.339	0.331	0.302	0.262	0.242	0.233
	taud		2.557	2.560	2.537	2.435	2.456	2.472	2.429	2.437	2.512	2.612	2.583	2.548
	Ebn at Noon		299	314	317	308	305	303	294	293	296	300	293	289
	Edh at Noon		20	24	29	34	35	34	36	34	29	23	20	19
(44) All-Sky Solar Radiation	RadAvg		664	991	1401	1748	2019	2389	2316	2012	1626	1100	736	539
	RadStd		45	60	77	101	172	142	91	64	108	75	55	35

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	-2.66	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page